

CrossPlayd

*Final Project for Course: Management of
Information Technology*

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1. Project Description and core functionalities

1.1 Project description

The purpose of the application **CrossPlayd** is to allow users to track and review video games they've played all in one place. Rather than being a replacement for other video game scoring sites, like MetaCritic or Steam Community Page, this application focuses on creating a user-based community for every game of interest, no matter if the user owns it and independently of the platform they've played it on. This also allows for logging of older or discontinued games, as well as games that are not available on major platforms.

1.2 Project Management Team

Student ID	First Name	Last Name
89231167	Aleksandra	Zec
89231086	Matej	Kunovski
89231138	Nikola	Severov
89231037	Andrej	Kaziovski

All members take on the project manager role, as well as the project client role

1.3 Links to project

Platform	Link
GitHub	https://github.com/aleksandrazec/CrossPlayd
Figma	https://www.figma.com/proto/HEsaUyWCWtv/zsnrnOBn5R/CrossPlayd?node-id=15-305&t=GmMJEkHxFzyGk50q-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=8%3A14
Application Demo	https://app2.stackrack.org/

1.4 Core functionalities

- Users can view a homepage which contains multiple tabs that display different video game categories, like trending, best rated, etc.
- Users can log video games they've played in their library and are able to rate them from 1 to 10.
- They can categorize the games by marking them with 100 Percented, Completed, Playing, Paused, Dropped and Plan To Play.
- Users can pin their favourite games on their profile.
- Users can leave comments on other users' accounts.
- Users can follow other users. In the case both follow each other, they have the status friends.
- Users can look through their following or friends' recent activities by checking the appropriate tab.
- Users can go to the forum page, where they can open new forums or comment on other users' forums.
- Users can inform themselves on video games they're interested in by reading the game description, reading the reviews made by other users and the global rating of the game.
- Users can interact with other users' reviews by liking and replying to them.
- Users can check out related games on the games' page they are currently on.

2. Personas and problem scenarios

We created three personas to help us better understand the problem that we are dealing with, and to get a clearer view of the implementation process. They provided a valuable perspective on how we should proceed with the development of the project and what our requirements should be.

2.1 Personas and problem scenarios

Marko Miler - The Hardcore Gamer

Screening question

How many games have you fully completed, unlocking all the trophies or achievements?

Description

Marko is a hardworking employee that really likes talking to customers and being helpful. He is a video game fanatic and spends a lot of time playing video games with his friends online. Over the years he has built a big collection of video games and consoles. Apart from that Marko also likes going to the gym and he attends anime and video game conventions regularly.

Demographic:

- Age: 28
- Gender: Male
- Marital Status: Single
- Income: €28,332 annually
- Occupation: Big Bang retail worker

Goals:

- Save up money so he can move out to his own apartment.
- Earn a raise and move up in the company.
- Finish all the games on his Bucket List.

Perspective

Perspective	Marko's Point Of View
Thinks	That his game wish-list is becoming never-ending and he can't keep track of it.

Perspective	Marko's Point Of View
Sees	His friends track their video game wish-lists on multiple different platforms.
Feels	Overwhelmed from the amount of new games that are coming out soon that he wants to log and play.
Does	Tries to be informed on new games by reading online articles and speaking with his friends.

Problem Scenarios

Problem Scenarios	Alternatives	Value proposition
Can't keep track of which games he wants to play and where they are available on	Uses a note application to keep track of them or makes multiple wishlists on different platforms	Keep track of all the games you want to play in the same place
Wants to be updated on what games his friend has finished	Ask his friend when he has the chance	He can visit his friend's profile and check his new updates
Wants to show off his gaming collection and all the games he finished	He shows off to his friends while playing with them	He can make a profile and log his gaming collection easily

User Stories

User Story	Test Cases
As a Hardcore Gamer, I want to log my video games in my library so that I can keep track of my collection.	- Make sure that you can add video games into your library. - Make sure that games can be sorted by different categories in your library.
As a Hardcore Gamer, I want to be able to see what my friends are playing so that I can keep up with them.	- Make sure that if two accounts follow each other they are classified as friends. - Make sure that you can see libraries of other accounts.

User Story	Test Cases
	- Make sure that there is a tab to view friend activity.
As a Hardcore Gamer, I want to express myself and my interests so that I can show off my collection to my friends.	- Make sure that you can set a favorite game to show up on your account. - Make sure that you can change your profile details.

Ellie Baker - The Gaming Youtuber

Screening question

How many gameplay videos have you made for your social media platform?

Description

Ellie Baker is a passionate gaming influencer that films lengthy gameplays of games her audience recommends to her. Because of her witty humor and problem-solving skills, her fans enjoy watching her content, since it feels like she's taking them on the adventure. Behind the cameras, she likes hanging out with her friends, watching anime and cuddling with her many pets.

Demographic:

- Age: 24
- Gender: Female
- Marital status: Single
- Income: \$140,000 annually
- Occupation: Youtuber

Goals:

- Reach 2 million subscribers on YouTube.
- Move to Los Angeles.
- Become a voice actor for video games.

Perspective

Perspective	Ellie's Point Of View
Thinks	Ellie is thinking about some way to connect with her fans further, and possibly reach new audiences.
Sees	Ellie notices that other youtubers are making content across many different social media platforms.
Feels	Ellie has liked playing and talking about video games ever since she was little. She gets fired up when fans want to discuss their views and understandings of the video games she's played.
Does	Ellie uploads gameplay videos on her channel almost everyday.

Problem scenarios

Problem Scenarios	Alternatives	Value proposition
Ellie needs a way to keep track of the games she's played for her channel and in her personal time.	She looks at her upload history on her channel, which is too large to be practical.	Having a personal tracker where she can put appropriate labels on each game she's played no matter the platform.
Ellie wants a way to connect further to her audience, outside of her videos.	She sometimes posts blogs or pictures of what she's doing in her free time.	Having a place to post her personal reviews and thoughts of the games she's played both on and off camera, would allow her audience to feel closer to her, and could discuss the topics of the game.
Ellie needs a way to see what games are currently trending, to know what she should play next for her channel.	She looks at what other gaming influencers are posting or looks up new games online.	Having a place to see what the community is currently reviewing and commenting

Problem Scenarios	Alternatives	Value proposition
		on would allow her to stay ahead of the curve.

User stories

User story	Test cases
As a gaming youtuber, I want to deepen my relationship with my fans, so that I can strengthen my online persona.	- Make sure that people can find and follow others by a username. - Make sure that people can like and reply to others reviews. - Make sure that there is a tab of most popular reviewers currently.
As a gaming youtuber, I want to know which games are currently popular, to ensure my content is relevant.	- Make sure that there is a tab of most reviewed games currently. - Make sure that there is a tab of highest reviewed games currently.
As a gaming youtuber, I want to keep track of my gaming habits, to ensure I don't forget to film specific gameplays.	- Make sure that the game library includes games across different platforms. - Make sure that the games in the library can be categorized with different labels.

Jane Garret - Lover of retro and arcade games

Screening Question

How many retro or arcade games have you played?

Description

Jane Garret is a married mother of two who works as a public defendant in a law firm. She got her law degree at New York University. She loves gardening, spending time with her family and origami. She used to play video games when she was a child and she grew up with arcade and retro games. She stopped playing video games as she got older.

Demographic:

- Age: 42
- Gender: Female
- Marital Status: Married
- Income: \$92,000 annually
- Occupation: Public Defendant

Goals:

- She wants to spend more time with her family, especially her children.
- She wants to try out new things and hobbies even though they might not work out.

Perspective

Perspective	Jane's Point Of View
Thinks	She liked video games as a child and wants to connect with her kids.
Sees	She sees there is no place to track what games you've played, especially retro/arcade games.
Feels	She feels an old nostalgic feeling to something she loved as a child. She wants to reconnect with her kids but also recapture that love she had for video games as a child.
Does	She watches them and spends time with her kids while they're playing video games.

Problem Scenarios and their Alternatives

Problem Scenarios	Alternatives	Value proposition
She wants to write down all the games she played as a kid.	She can physically write them in a notebook or send them in a message to her children in a group chat. This is tedious and impractical.	Having a platform where she can track her games and that being easily viewable would solve her issue.
She wants to see what games her kids are into.	She can walk into their room and look through their gaming	If there was a platform where she can follow her kids and

Problem Scenarios	Alternatives	Value proposition
	collection or she can ask them.	can track what games they've played, that would solve her problems

User stories

User story	Test cases
As a mother of two children, I would like to spend more time with my family and do some fun activities together. I would like to try and find a couple of party games for the whole family that keeps everyone engaged and is interested.	- Make sure that the platform provides search options by genre and user preference. - Make sure that the platform gives the most popular options when searching.
As a former retro gamer, I would like to replay some of my favourite games from the past and see if my beloved game series have some new additions to try out.	- Make sure that the platform provides a wide catalogue of games from past and present. - Make sure that the platform is up to date with the newest sequels or other games linked to a game series (Example: Final fantasy series)
As a former gamer that wants to rekindle my spark for video games again, I would like to be able to see opinions and reviews of my fellow gamer buddies.	- Make sure that the platform provides a review system. - Make sure that the review system lets you reply and like reviews/comments provided by the users.

3. Hypothesis testing

After creating our desired personas, we moved forward with our hypothesis testing. This is a vital step in our requirements definition, which would solidify that our application is being developed with the right focus and ideas in mind.

Our hypothesis states the following: *The average casual gamer would like to track and or record their game library in one single place. They also value reviews and the social aspect of gaming.* We constructed various questions based on it.

3.1 Introduction

We interviewed a group of 5 Computer Science students who play video games and are aged 21 to 22. interviewees are defined below:

- Student A - Stefan Radojichikj
- Student B - Gorazd Gajdov
- Student C - Nikola Rupnik
- Student D - Filip Trajkoski
- Student E - Vedran Gvozderac

3.2 Hypothesis:

The average casual gamer would like to track/record their game library in one single place. They also value reviews and the social aspect of gaming .

Questions:

1. Approximately how many hours do you play video games in a week?

- A: It would vary depending on the situation, but roughly around 10-12 hours in a week.
- B: I usually play around 15 hours a week.
- C: Rough estimate is around 14 to 16 a week.
- D: Roughly 1 to 2 hours.
- E: Like 3 hours a day so 21 hours weekly.

Conclusion: We have interviewees with different gaming habits.

2. Do you play video games on multiple platforms? Which online store do you utilize?

- A: No, I only play on PC. Usually Steam or Epic Games Store.
- B: Yes, I do. I play both on PC and PlayStation. I usually use Steam on my PC.
- C: No, I only play it on PC. Mainly Steam.
- D: I play on PC mostly. I use Steam and Epic Games Store.
- E: If mobile counts, then 2. I use the App Store on mobile and Steam on my PC.

Conclusion: Even though most people play on their personal computers, a majority of them use multiple platforms and stores to play and purchase video games.

3. Do you track your gaming library and if so how do you organize it?

- A: I don't really take any initiative to track my gaming library. I usually keep it updated through Steam with the built-in feature.
- B: I don't organize my library by myself. I have my games on Epic and Steam libraries.
- C: I have almost everything on Steam so it does it automatically I guess.
- D: Not regularly, nothing beyond Steam and Epic Games.
- E: I don't organize it, but I do track my Steam games and achievements a lot. Big fan of that.

Conclusion: They do not organize their gaming libraries by themselves and usually use platforms that do that for them.

4. How do you inform yourself about any upcoming games?

- A: Word of mouth, YouTube, online platforms, or friend recommendations.
- B: Mainly Instagram and YouTube.
- C: Usually just see videos online and on social media.
- D: From social media, Instagram and YouTube.
- E: Instagram and friends.

Conclusion: There are a lot of ways people inform themselves on new video game releases and they vary from person to person.

5. How important are video game reviews for whether or not you will play the game?

- A: Not too much. I just use them as an introductory source of information for whether I would maybe like the game or to know how long it takes to beat it.
- B: Depends, sometimes I don't wanna watch reviews because I want to go in blind into a game. If they are generally positive, then I'll just go blindly into the game.
- C: Depends on the reviewer, if it's a good reviewer, then a lot, otherwise not at all.
- D: They are important, but I would also like to try the games myself to see.
- E: Not that important. The number is not that important but I like reading the individual reviews a lot.

Conclusion: Even though people don't really see the ratings, they still care about the general opinion of players and like to read their reviews to inform themselves.

6. Where do you find video game recommendations? (i.e. new video games to play)

- A: Word of mouth or friend recommendations.
- B: Either from friends, YouTube or if I see a trending game on Instagram or TikTok.
- C: Usually I get recommended games by friends.
- D: Mostly from YouTube.
- E: Someone recommends it to me and I play it I guess.

Conclusion: Usually most people like to play games that are recommended to them by friends.

7. Would you use a platform for tracking your whole game library in one place?

- A: Yeah, yes I would.
- B: Absolutely, I would definitely use a video game library tracker.
- C: Well, I already use Steam, so no.
- D: Yea, sometimes it gets annoying to try to find on which platform I own the game.
- E: Yes, but it would be cool if it would also track my hours.

Conclusion: Most people want to organize and keep track of their libraries in one place.

3.3 Conclusion

The hypothesis is viable. This means that it is true that the average casual gamer would like to track and or record their game library in one single place, as well as that they also value reviews and the social aspect of gaming.

4. Usability testing

With our new gained knowledge from the hypothesis testing, and a better understanding of user wants and needs, we constructed a figma prototype of how our application should look like. We then tested our prototype to a number of users. In the tests we used our Figma Prototype with hardcoded values.

4.1 Assessment Testing

Objectives and Methods

Our objectives are defined as:

- Understand if something confuses users while using our prototype
- Test alternative solutions
- Understand user needs and requirements

The testing method that was used: moderated, in-person and qualitative.

Subjects and Screening Questions

We will choose subjects based on these two screening questions:

- Do you play video games regularly?
- Do you read online video game forums and reviews?

Subjects:

- User 1 - Eva Petrushevska
- User 2 - Simona Cholakova
- User 3 - Ana Milenkoska
- User 4 - Stefan Radojichikj

Research Composition

Item	Duration (min.)	Notes
Intro	5	Explaining the objectives of the test
Test Tasks	10	Introducing test scenario and completing tests
Post-Test debrief	5	Additional questions about prototype

Defining Tasks

Tasks for Assessment Testing:

- Task 1 - Log into the application.
- Task 2 - Navigate to Games Tab.
- Task 3 - Add Resident Evil Requiem to your Library with Status Playing and Rating of 8 stars.
- Task 4 - Add a review to Resident Evil Requiem.
- Task 5 - View Resident Evil Requiem in your Library.
- Task 6 - Navigate to the Community Tab.

The starting point of Task 1 is the Home page of our application. After the first task, all subsequent tasks start subsequently, and each starting point is the last screen in the previous test.

Notes on Execution of Testing

What would need to be fixed in the solution?

- User 1 - Had no problems with the solutions.
- User 2 - Said that the library button was too small in Task 5
- User 3 - Said that the login text was too small in Task 1.
- User 4 - Thought everything was fine.

Did any of the tasks cause particular problems?

- User 1 - No, did not have any problems while testing.
- User 2 - Was a little confused while doing Task 3 but did it in a timely manner.
- User 3 - No, did not have any issues while testing.
- User 4 - No, didn't have any issues.

How much time did they spend on each task (sec.)?

User	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6
User 1	10	4	12	10	7	3
User 2	15	3	23	13	11	2
User 3	8	3	13	5	12	3
User 4	7	2	11	10	12	2

Which task was most intuitive for them?

- User 1 - Task 5 was the most intuitive.
- User 2 - Task 6 and Task 4 were the most intuitive.
- User 3 - Task 5 and Task 6 were the most intuitive.
- User 4 - Task 6 was the most intuitive.

Post-Test Debrief

Additional Comments from users:

- User 1 - Thought that the ratings should be switched and presented from lowest to highest in Task 3.
- User 2 - Said that the search functionality, which was not fully implemented in the prototype, would make it easier to search for games.
- User 3 - Thought that the design was nice and the features were well implemented.
- User 4 - Said the layout of the account page was a little confusing and it would be better if it was more intuitive.

4.2 Conclusion

From this testing, we learned what features should be adjusted and which functionalities were intuitive to the users and which weren't. With this newfound knowledge, we proceeded with the development of the application, considering the user's needs.

5. Project stakeholders

For the purpose of being sure about what risks and uncertainties were present for the making of this project, we conducted an analysis on all of the project stakeholders. This included describing the attributes regarding each stakeholder, as well as benefits and conflicts that might appear. This helped plan ahead for the possible difficulties that might arise during the project's lifetime, as well as play into the project's strengths.

5.1 Stakeholder: Team members

Type of Stakeholder: Development Team

Status: Positive

Expectations/Interests: It's in our interest if our application succeeds, even if our expectations are more level headed

Conflicts: The development team is composed of students and since this is a passion project, there is no pay. There are possible time conflicts between student obligations and development obligations

Benefits: We benefit massively in our professional careers, if our applications succeeds. It is a great addition to our project portfolio

Team response: Approaching the development in an incremental approach, with a digestible workload each week and strong communication, so the team members don't feel unmotivated or overwhelmed by the obligations

5.2 Stakeholder: Gamers

Type of Stakeholder: End users

Status: Positive

Expectations/Interests: The expectations for the application is a user-based community for reviewing, rating and tracking video games from different platforms

Conflicts: Conflicts could arise in the case that gamers prefer some other way of tracking games they've played, they play games only on one platform or they are satisfied with the current game reviewing platforms.

Benefits: The more gamers use the application, the better the user experience is, since there is a larger community, meaning more reviews, discussions and information available. The benefit is the exchange of knowledge between users

Team response: We will focus on the user experience and ensure broad functionality for our application, to make it more appealing. We will conduct user testing during development, and adjust our application based on the gamers' responses

5.3 Stakeholder: Game Critics

Type of Stakeholder:End users

Status: Neutral

Expectations/Interests: It's in their interest if our app succeeds because interest and interaction with game reviews and discussions will surge

Conflicts: There might be conflicts if they are sponsored by one of our competitors, like for example, Steam

Benefits: They benefit if our application is popular and if there are more people in the community. They can close the distance between real users and professional critics by having a profile on our application

Team response: We believe that the service we provide is better for game critics because our application encourages more community interaction and focuses on promoting game reviews, even if we do not sponsor them. To incentivize game critics, we will implement verified profiles for them and promote their reviews on our application

5.4 Stakeholder: IGDB.com

Type of Stakeholder: API Provider

Status: Positive

Expectations/Interests: The API providers, IGDB.com, want attribution for the use of their API for commercial use

Conflicts: There are no major conflicts

Benefits: IGDB.com get free exposure to their website and API

Team response: We will add clear attribution and we will link IGDB.com in the About Us page.

5.5 Stakeholder: Steam

Type of Stakeholder: Competitors

Status: Negative

Expectations/Interests: The success of our application is not in Steam's interest, since the use and function of our application is broader for the task of game tracking.

Conflicts: The user community on Steam is threatened by our application, since we provide a larger framework for user interaction

Benefits: More reviews from our website leads to more Steam game sales

Team response: Steam's library can get cluttered over time and it becomes very tedious to navigate. We will implement a more smooth and efficient library tracking experience, with a bigger assortment.

5.6 Stakeholder: Data regulatory bodies

Type of Stakeholder: Government

Status: Neutral

Expectations/Interests: Follow the guidelines and laws about user data as stated by the government

Conflicts: If we don't follow the law, we would be given a fine.

Benefits: No benefit for the data regulatory bodies, however, users may feel more secure and trusting of our application if we ensure data protection

Team response: We will follow the rules and guidelines of user data protection, as well as ensure secure authentication and database interaction

6. Scrum basics

We chose to implement an agile project management approach, since we are a small team of developers and first time project managers. We require an approach that is efficient in its communication and works in an incremental manner, so that our progress can be tracked and managed accordingly. For this purpose, we chose to implement the SCRUM methodology, using Kanban boards. We implemented four sprints, each of which lasted approximately two weeks. This approach proved beneficial to us and played to our strengths as a small, new team.

6.1 Sprint 0 Information

Team roles

Student ID	First Name	Last Name	Role
89231167	Aleksandra	Zec	Scrum Master
89231086	Matej	Kunovski	Developer
89231138	Nikola	Severov	Designer
89231037	Andrej	Kaziovski	Developer

All of the team members also take on the Product Owner role.

Technologies

- Prototype design
 - Figma
 - Affinity
- Front-end
 - React
 - JavaScript
 - HTML & CSS
- Back-end
 - NodeJS
 - Express
- Database
 - Supabase
 - PostgreSQL
- API
 - IGDB API
 - <https://api-docs.igdb.com/#getting-started>

- Deployment
 - Vercel
- DevOps and Tools
 - Visual Studio Code
 - GitHub
 - Obsidian
 - Postman

Report

In the sprint 0, the following tasks were finished successfully:

- Core functionalities were defined
- Budget estimate was defined
- Personas and problem scenarios were created
- User stories and test cases were defined
- Hypothesis Testing was conducted
- Low Fidelity Prototype was created
- Usability Testing was conducted

6.2 Sprint 1

This Sprint was mainly focused on the setup of the application environment. This was the foundation that made other sprints possible. Despite the Sprint tasks weren't defined properly, we managed to continue in a proper manor. All of the tasks were completed.

Completed

- ☒ Create branching strategy
- ☒ Create React project
- ☒ Set up database
- ☒ Set up account for API
- ☒ Create NodeJS & Express back-end
- ☒ Create basic queries from back-end to API

6.3 Sprint 2

In this sprint we started developing the functionalities of the application. We had some difficulties and had to delay some of the tasks. Despite that, all of the work was completed and tasks finished.

Completed

- ☒ Create Log In/User functionalities

- ☒ Create Forum functionality
- ☒ Create Game's page
- ☒ Create Games Tab

6.4 Sprint 3

The third sprint of development was the last sprint before the presentation of the application. This meant that we wanted to implement all the functionalities needed to have a working demo. In the process we had to cancel one of the tasks.

Completed

- ☒ Make session data consistent
- ☒ Add library functionality
- ☒ Add review/star functionality

Cancelled

- ☐ ~~Add follower following~~

6.5 Conclusion

Even though the development of the full application isn't finished, these four sprints allowed us to get the application to a point where it was presentable and had enough functionalities to represent our ideas and values properly. During the period of the sprints, not only did we deliver a working demo, but we also made proper documentation and managed our workload successfully.

7. Team patterns

In the interest of bettering team dynamics and bettering morale, the following is an analysis on our team's positive and negative patterns.

7.1 Team's negative patterns

Pattern 1: Ignoring low performers

A pattern that was prominent at the start of development of the application was ignoring low performers. Not everyone in the team was equally as familiar with the technology being used, which resulted in team members having variable efficiency for the tasks assigned to them. This meant that deadlines for certain tasks weren't being met and the development had unrealistic timelines set.

To battle this, the team is meeting in real life at least once a week to work and develop together. While each member still has individual tasks assigned to them, such an approach means that the team members are able to discuss issues that might arise and assist each other in difficult areas. We also adjust the deadlines and goals appropriately for each team member, so everyone is working on their tasks in a realistic and timely manner.

Pattern 2: Being everyone's friend

Considering the fact that the team members are classmates and friends, there is a pattern of trying to avoid any sort of conflict to make sure that the private relationships are not affected by the project's development and success. However, this means that when some parts of the product weren't developed with the proper care, the members are afraid to critique the work, discuss the issues explicitly and confront each other about it.

To tackle this issue, we make sure that critiques pointed at each other's work are as objective as possible, and aren't framed as personal attacks towards the members. Also, we make sure to draw a clear line between the dynamics of members during development and outside of the work, so that the issues don't transfer between the areas, and impact our personal relationship, nor compromise the team dynamics.

Pattern 3: Technical Perfectionist

As ambitious and motivated individuals, we sometimes find it hard to not be perfectionists during production. Balancing quality, deadlines and our personal satisfaction with the product is a complex task that requires a lot of focus and clarity during development, when it's easy to get lost in the tasks of the job.

To assure the team is moving forward with the tasks at hand and the deadlines are still being met, we will try to define the tasks in a concrete and grounded manner. This means that the goals set should be specific, incremental and measurable, and there should be really explicit

expectations for each upcoming milestone. This way, the members are sure that they aren't overachieving or spending too much time on perfecting their work.

7.2 Team's positive patterns

Pattern 1: Be a Teacher and a Mentor

A positive thing that the team members implement during work, is helping each other in a proper manner. When a team member is confused or stuck on some part of their task, to help them, another member comes and guides them in the right direction, providing key information or resources needed for the solution. In this situation, the members don't try to overexplain things or ramble for too long, and instead give the one asking for help space to figure it out themselves, and let them know they are available if additional clarification or help is needed.

Pattern 2: Be a Catalyst

In an effort to work efficiently and in agreement, team members often engage in productive conversations and discussions. When a team member seems to take lead for a particular part of the product, the others don't mind following them and discussing directly with them for the purpose of reaching common goals and consensus. Depending on the functionality/part of the product, different members are assumed for this role, depending on their expertise and passion for the task at hand.

Pattern 3: Track Happiness

To keep morale up, the team members try to maintain an upbeat and low-stress environment during work. During work, the team members keep each other's needs in mind and lift each other up. Meetings and productive conversations are held with consideration of the members temperament, mood and their current needs. Being happy and content while working means being more productive and providing higher quality solutions, so we don't neglect this even in higher stake situations.

7.3 Pattern Conclusion

Even though the team has some anti-patterns that might be compromising the efficiency of production, the morale is kept high and we try to battle the issues present at the moment, and are constantly improving. Considering the team is relatively new and inexperienced, as well as the fact the team dynamics are still being concretely defined and established, we find that things are moving along at a nice and healthy pace.

8. Conclusions

This was our first significant project as a team, where each member took on a role both as a project manager and a developer. There was a significant difference between development of a project as a solo developer and managing and developing a project as a team. From the early stages of development with the creation of the personas, followed by usability testing and all the way up to the final application demo as the end goal, we adapted to the challenges of managing and developing a project alongside each other.

Throughout the making of the project there were a lot of difficulties and roadblocks that halted our progress. Specifically, the making of the demo posed some great difficulties that we had to overcome. Despite these issues, we continued progressing and finished the project, even if we had to cut some functionalities along the way. While developing the project, we had to narrow the scope because of technical difficulties and time constraints. Even if we did not clear all of our original goals, we are pleased with the end product and these cut functionalities can be set as future goals when we decide to return to this project.

Working on this project as a team has proven to be an immensely valuable and rewarding experience. Moreover, it helped us learn the pipeline of development of projects and their management, which will be applicable further in our careers.

9. Assessment of work

9.1 Assessment of work by Matej Kunovski

I assess that the work was done equally between all team members. All team members actively contributed while we were doing the homework assignments and planning of the prototypes or demos. As for development of the application: I, Aleksandra Zec and Nikola Severov performed most of the work for the application. While on the other hand, Andrej Kaziovski worked on the reviews functionality and frontend for the application and did the documentation and presentation. I would also mention that the team was usually split into smaller teams of two for doing different tasks while programming or in the homeworks. Lastly, I think that all of the team members contributed equally.

9.2 Assessment of work by Aleksandra Zec

I think that the division of work during the development of this project was fair. All of us participated and worked on the weekly homeworks together. We wrote most of them all together, with the exception of the testing homeworks. Matej and Andrej worked on the testing while Nikola and I worked on the Figma prototype. Later on, when we started working on the application, we planned its structure and Andrej and Nikola set up the database, while Matej and I set up the framework for the frontend and backend. Even though all of us worked on the development, perhaps Matej, Nikola and I focused more on the coding and Andrej more on the documentation. However, I think that everybody contributed equally and that this division worked well for our team.

9.3 Assessment of work by Nikola Severov

For the Homeworks, we would usually group up in the same place, and do the homework together. If one of us could not make it, or wasn't able to work that day, the three remaining members would still meet and do the work. All of us took one of these days, still making everyone do the same amount of work. For presenting, we would rotate who was presenting that week. For the Implementation of the project, most of the work was equally split by me, Aleksandra and Matej and Andrej also worked on the implementation but less than us three. But to make up for that, he did the Documentation for the project, and he made the presentation for the project. In my opinion, the work was evenly split, and every member contributed equally to the overall project.

9.4 Assessment of work by Andrej Kaziovski

I think that the work was distributed equally except for the prototype, where my colleagues did most of the work except that I helped with the ReviewBox section of the application as well as with the frontend, documentation and the presentation. On the other hand following the homework assignments we would usually group together or split into teams of two, where we would finish our respective assignments. Me and Matej did the User and Hypothesis testing while Aleksandra and Nikola made the Figma prototype. I think that the work was evenly distributed for the homeworks, while for the final prototype implementation it was not as mentioned before that my colleagues did most of the work.

10. Showcase of application

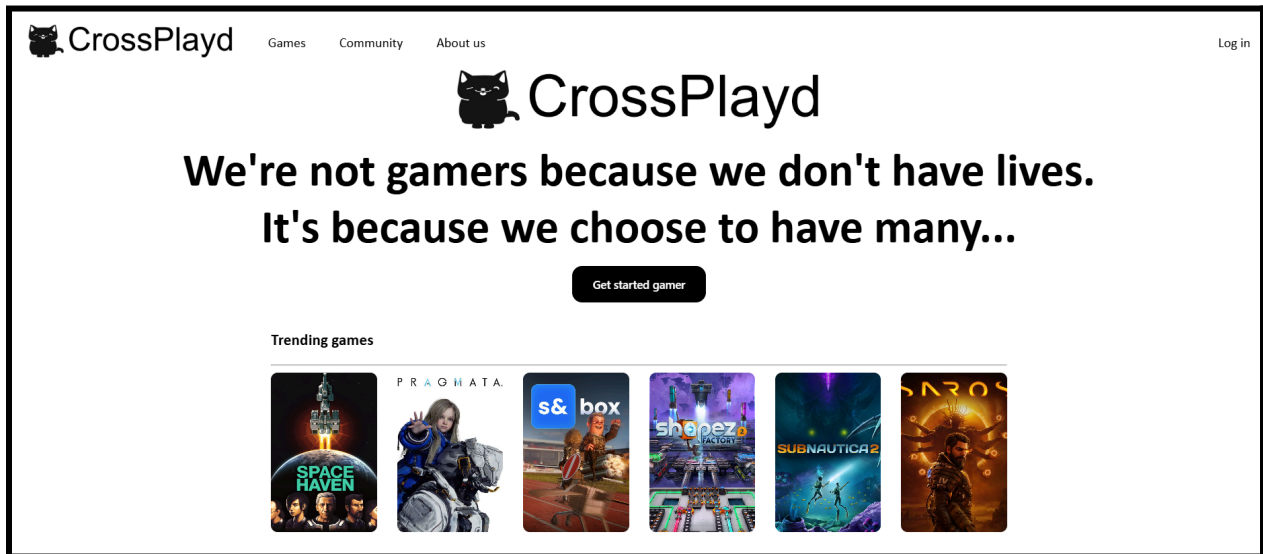


Figure 1 - Home Page

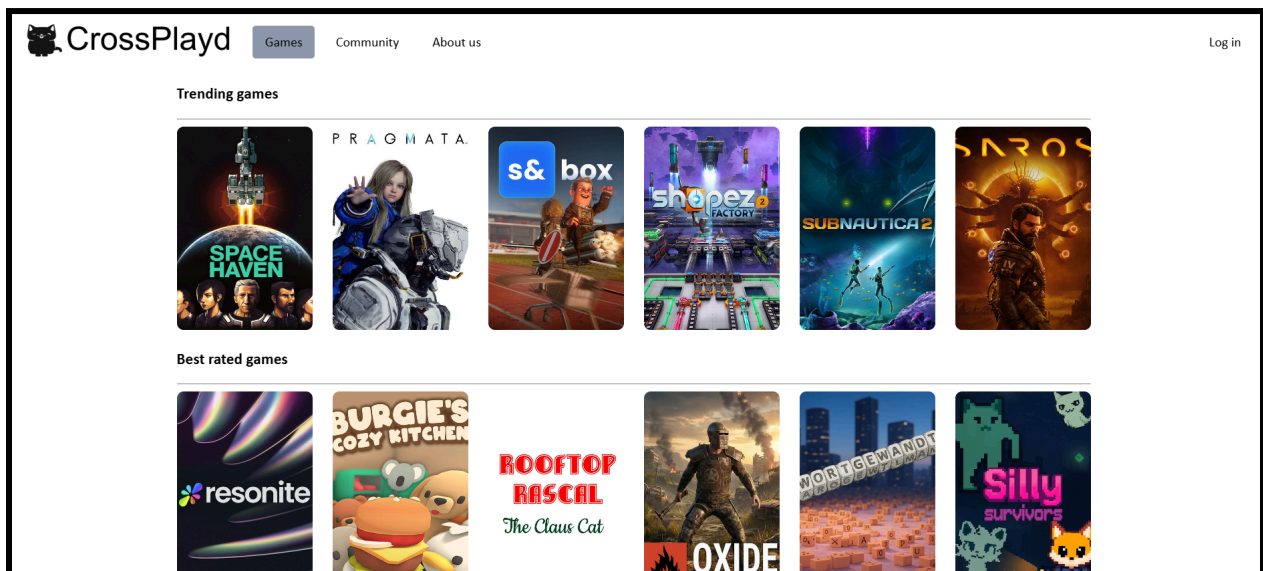


Figure 2 - Games Page



Figure 3 - Login Page



Figure 4 - Profile Page

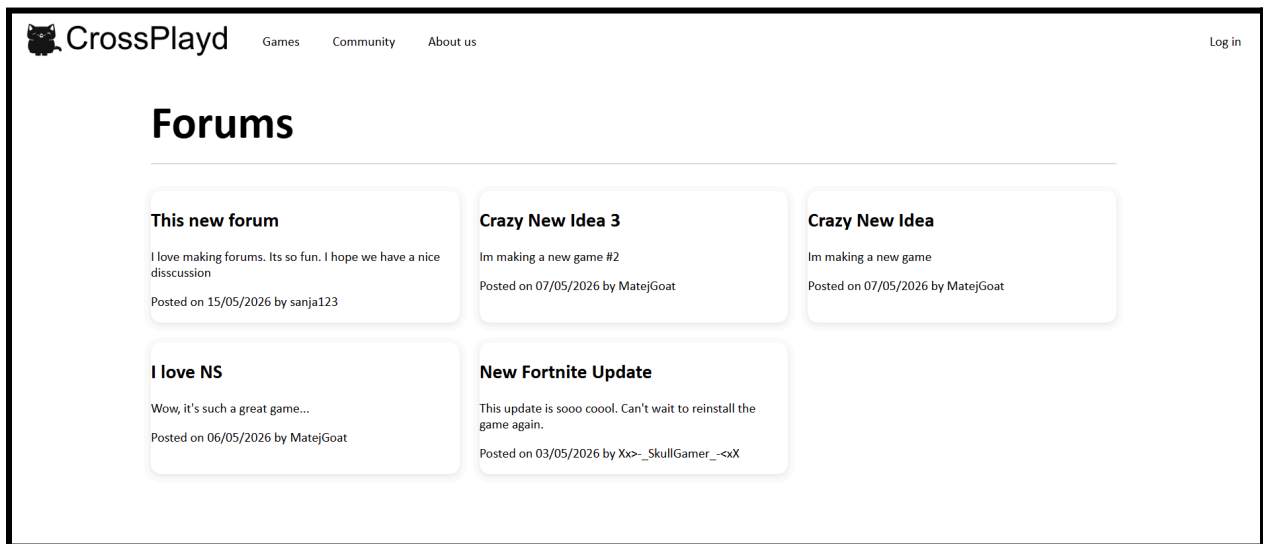


Figure 5 - Forums Page

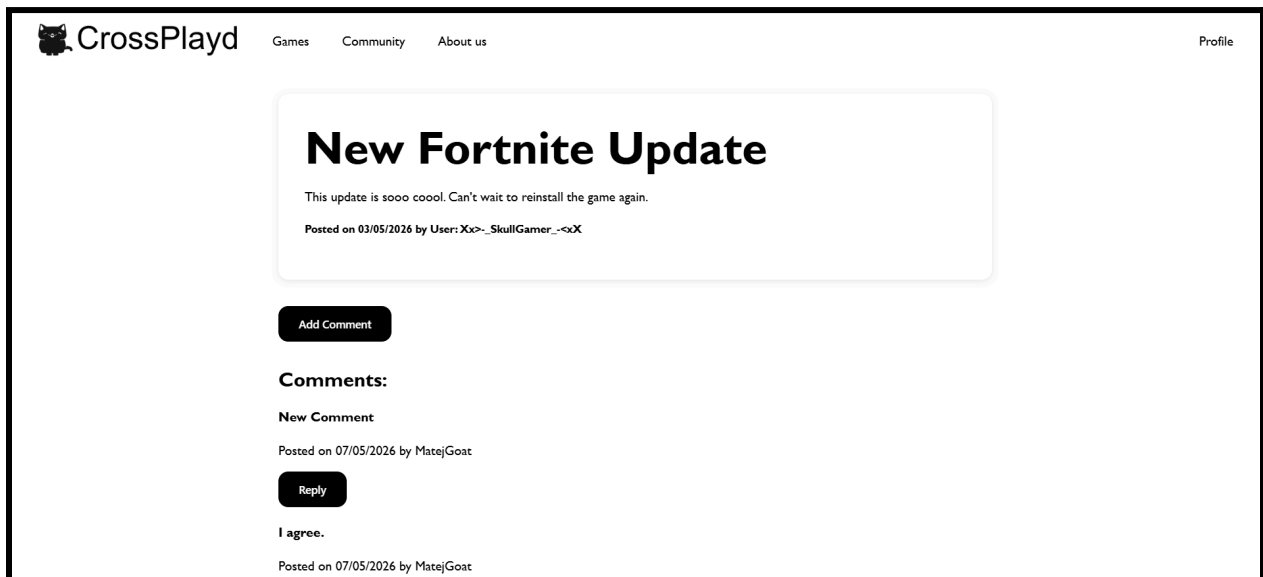


Figure 6 - Page for Specific Forum

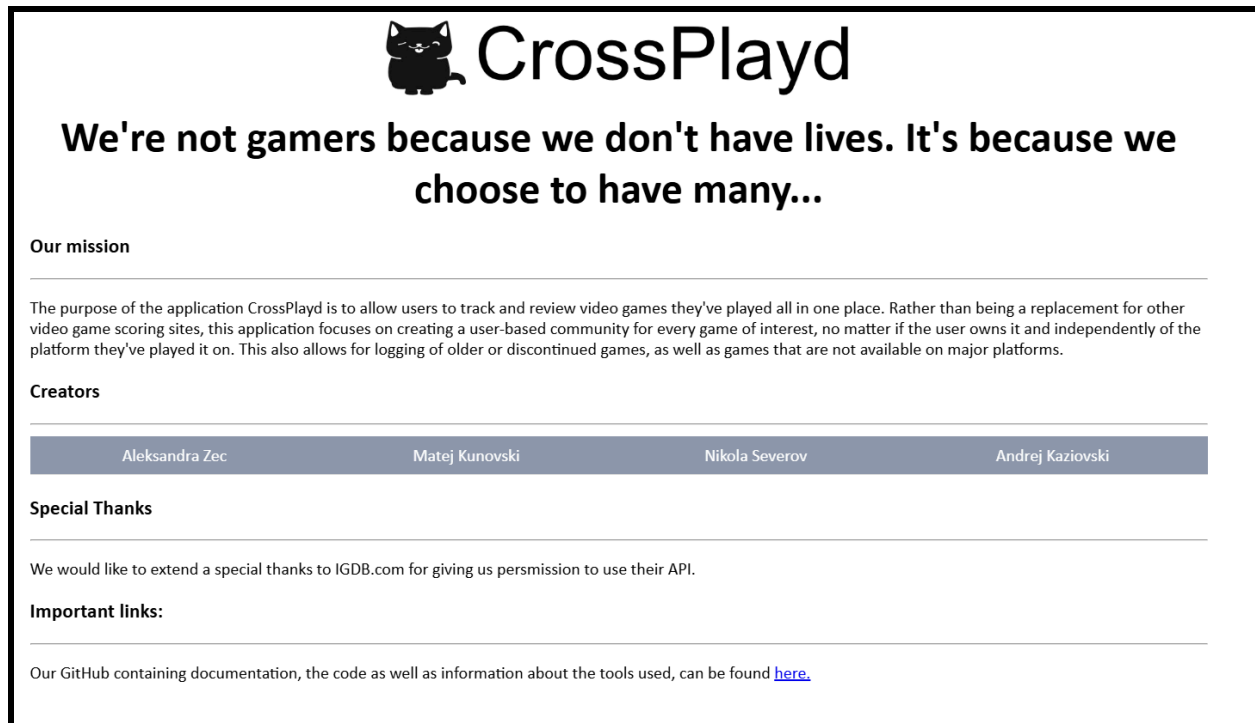


Figure 7. - About us Page

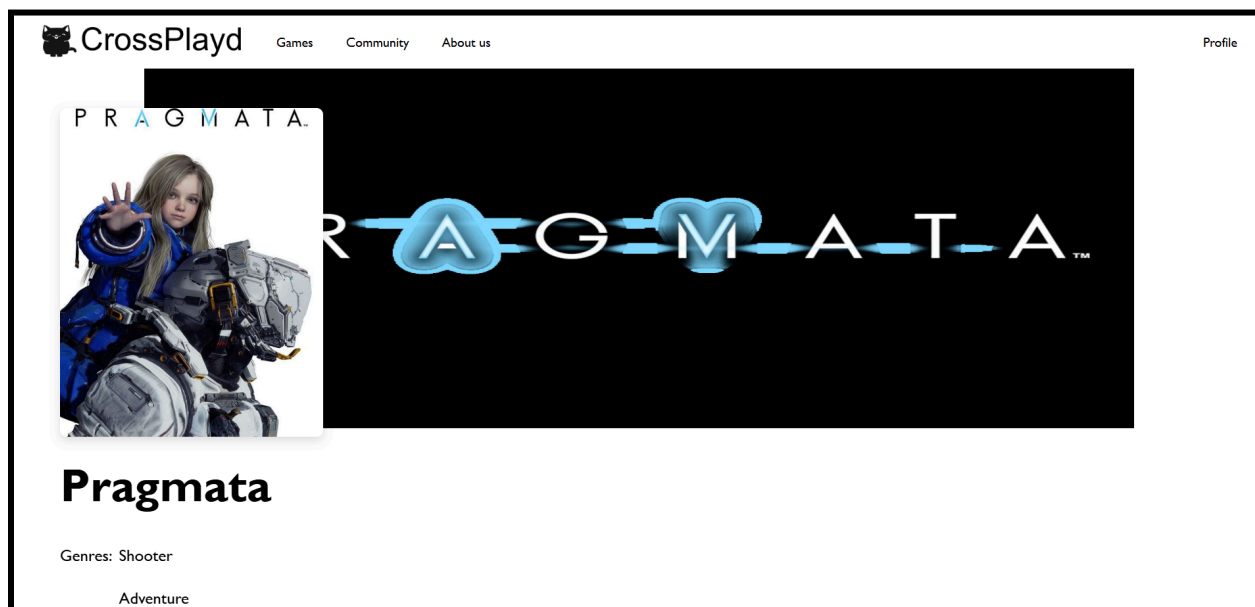
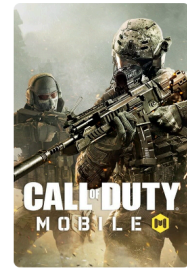
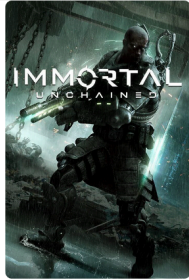


Figure 8 - Page for Specific Game #1

Pragmata is a science fiction action-adventure game set in a near-future environment. The story follows Hugh and his android companion Diana as they navigate a lunar research station. Gameplay incorporates action elements alongside mechanics involving interaction with systems within the environment.

Related games



Your Rating: 3

Edit Library

Figure 9 - Page for Specific Game #2

Reviews

Add Review

I hate this game
Posted on 14/05/2026 by MatejGoat

WOOOOOW, crazy
Posted on 15/05/2026 by

Review #1
Posted on 15/05/2026 by MatejGoat

Figure 10. - Page for Specific Game #3

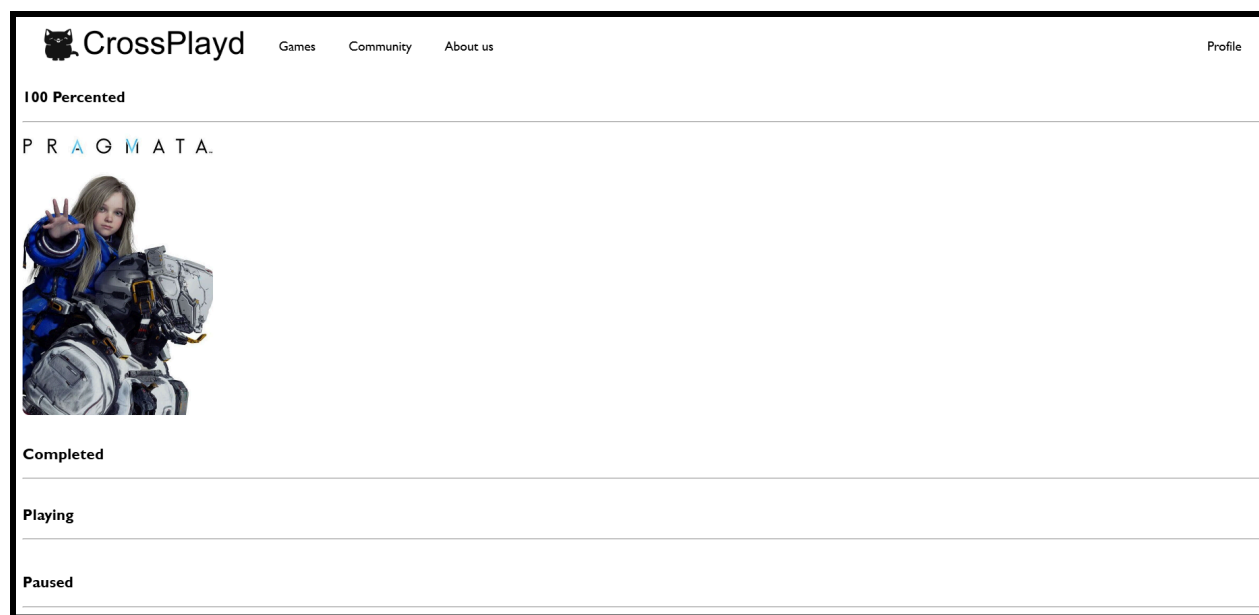


Figure 11 - Library Page